Media Effects or More Complicated? Examining the Relationship Between Online Media Fractionalization and Political Polarization

Lama Ammane

1009413332

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Professor Besco

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Introduction

Political polarization is a pernicious force that has the power to strangle modern states by undermining the legitimacy of the government. Scholars have long grappled with identifying a prominent underlying driver behind the rise in polarization in nations, both autocratic and democratic (that should in theory uphold the principles of plurality of ideals, and peaceful deliberation between citizens). While there is no clear consensus on a single force behind polarization, numerous scholars have identified media fragmentation as a prominent source. Media fragmentation is the degree to which the media is fragmented in its reports of political events, indicating different ideological factions. Political polarization is the level of hostility and ideological distance between political camps (Coppedge et al. 2024).

Scholars such as Kang, Kim and Lee (2025), have found that polarization has many negative outcomes on a society. Political polarization can affect economic outcomes by creating barriers to investment, cause economic uncertainty, and can lead to democratic backsliding since voters are highly divided. It may even cause voters to prioritize partisan interests (such as winning elections) above democratic principles, which is dangerous to the health and survival of democracies worldwide. Polarization can threaten democratic institutions and reduce the capacity of a state to cohesively address issues such as climate change and rising inequality (Törnberg 2022, 1). It is arguable that even in autocracies, political polarization among citizens can destabilize state control and threaten stability, potentially sparking violent protests or government overthrow.

Theory and Hypothesis

This paper hypothesizes that media fragmentation is the largest driver of political polarization in democratic states, and a significant (though not the largest) cause, in autocratic

states. Autocratic states may have historic features, such as colonial legacies, that also drive ideological fragmentation along political lines, hence media is not hypothesized to be a primary driver of polarization. This hypothesis was informed by the work of Tokita et al. (2021) who find that a fractured online media landscape leads people to self-sort into social networks who reflect their beliefs, thus driving higher polarization. Their analysis found that individuals who followed more polarized news outlets lost social ties with people from different ideologies at statistically significant rates, suggesting they intentionally pursued politically aligned social networks.

This paper attempts to answer the question "how is media fragmentation related to political polarization among different groups in a nation?" Does the media have a large and statistically significant correlation with polarization, holding other characteristics of a state or population constant? This paper will use descriptive statistics and a multiple regression model, one with and one without an interaction with regime type, to answer this question. The main hypothesis of this paper is that yes, media fragmentation has a significant relationship with polarization and may possibly cause it, holding other factors fixed.

> **H1**: Higher media fractionalization is associated with higher political polarization in a nation.
> **H2**: Higher media fractionalization is not associated with a change in political polarization in a nation.
>
> **H3**: Higher media fractionalization is associated with an increase in political polarization in a nation, holding GDP per capita, regime type, internet censorship, and head of state appointment type constant.
>
> **H4**: Higher media fractionalization is not associated with an increase in political polarization in a nation, holding GDP per capita, regime type, internet censorship, and head of state appointment type constant.

Interaction:

> **H5**: Higher media fractionalization in a democratic nation is associated with higher political polarization, whereas it is associated with ambiguous effects or no change in

polarization in autocracies, holding GDP per capita, regime type, internet censorship, and head of state appointment type constant.

H6: Higher media fractionalization in a democratic nation is not associated with higher political polarization, and is associated with ambiguous effects or no change in polarization in autocracies, GDP per capita, regime type, internet censorship, and head of state appointment type constant.

The larger theory behind this paper, informed by the academic literature, is that when online news media present different accounts for political events, viewers will sort themselves into the account that most aligns with their perceptions of the world, even if it does not perfectly reflect their own views. This ideological sorting causes people to cling more tightly to their perceived version of events. Over time, sorting leads to stronger feelings of animosity toward other groups and can escalate to hostility, while reducing willingness to engage in conversation peacefully with those with differing views. Hence, this paper will investigate whether online media fragmentation causes a breakdown of social cohesion in society and thus a rise in affective political polarization.

Dataset and Variables

This paper uses data from the Varieties of Democracy Dataset, focusing on the year 2019 with the filter command. After dropping all countries with unavailable data for a given variable (using the is.na command) the sample size was 113**.** V-Dem data is based on a survey of 4200 experts, such as academics and professionals, and the unit of analysis is countries.

The independent variable examined in this paper is "Online media fractionalization (**v2smmefra**)," which is posed as "Do the major domestic online media outlets give a similar presentation of major (political) news?" This is an ordinal variable, with responses organized in a 5-point scale, ranging from 0 (no, there is high variation in how major events are presented), to 4 (yes, online media largely present similar accounts of events). It is converted into an interval

level variable, making the data continuous and numeric. The dependent variable is political polarization (**v2cacamps**), posed as "Is society polarized into antagonistic, political camps? " This variable is ordinal, but is converted into an interval level variable. The responses range from 0 (no, different political camps interact in a friendly way), to 4 (yes, different political camps interact with hostility).

In addition, a variety of control variables were selected based on their theorized significance, or importance according to scholars. The purpose of adding control variables is to mitigate endogeneity (the correlation of another unobserved factor with the independent variable), such as omitted variable bias (an unseen variable affects both the independent and dependent variables). By including the following controls, this paper aims to more accurately model the relationship between polarization and fragmented media landscapes. The first control variable is GDP per capita (**e_gdppc**), an interval-level variable. This control was chosen because it may confound with media fragmentation by providing more funding for a diverse array of online news media. It may also correlate with political polarization because low economic development can precipitate social tensions. Hunter (2025) finds that greater state wealth and media consumption are associated with higher political polarization due to the increased prevalence in differing ideologies. The second control is internet censorship by the government (**v2mecenefi**), which is an interval-level variable. It was chosen because it may drive both media fragmentation, as well as political polarization. Censorship may potentially wall off certain viewpoints, contributing to different ideological camps, and further entrench existing ideological differences (increasing polarization). It is important to control for this variable because it may interfere with the relationship between media and polarization by creating different news perspectives online, as well as further deepening ideological divides.

The third control variable is regime type (**v2x_regime**). This is an ordinal-level variable, ranging from 0: closed autocracy, 1: electoral autocracy, 2: electoral democracy, 3: liberal democracy. It will be included as an interaction variable: a variable that alters the impact of media fragmentation on political polarization. Autocracies often restrict internet use and utilize censorship to filter content (Schleffer and Miller 2021, 95), which may lead to different effects than in liberal democracies, where internet freedom is high.

The fourth control variable is head of state appointment (**v2expathhg**). This variable is nominal, with 10 categories ranging from "appointed through violent force, such as rebellion", to "appointed through popular election". This was chosen because younger democracies tend to be politically unstable and prone to polarization, and thus the way the current head of state gained power can be a proxy for regime age and stability. This variable may impact polarization, as well as media fragmentation if the regime pursues populist means to maintain power, which can drive polarization (Schleffer and Miller 2021, 91). As this variable is nominal, it is initially transformed with the as.factor command so R will treat it as categorical.

Methods

This paper will examine the central tendency of political polarization, media fractionalization, and all the control variables, using their means or medians. It will also look at dispersion through standard deviation or interquartile range to interpret the spread of the data. Histograms of political polarization and media fractionalization across all countries will also be used to interpret the dispersion of the data visually (as they are interval variables). Lastly, a scatterplot between media fractionalization and polarization will be used to visualize the linear relationship between the independent and dependent variables, as well as a boxplot showing the spread of polarization across regime types.

Lastly, this paper will run a regression of polarization on media fragmentation (model 1). It will then create a second model using the following controls: GDP per capita, internet censorship, head of state appointment, and lower house electoral system, and examine whether this more robust model better predicts the influence of polarization on media fragmentation (through changes in regression coefficients or statistical significance). The final regression model will introduce the regime type variable through an interaction, and interpret whether regime type affects the association of media fractionalization with polarization, holding all the controls constant.

Results

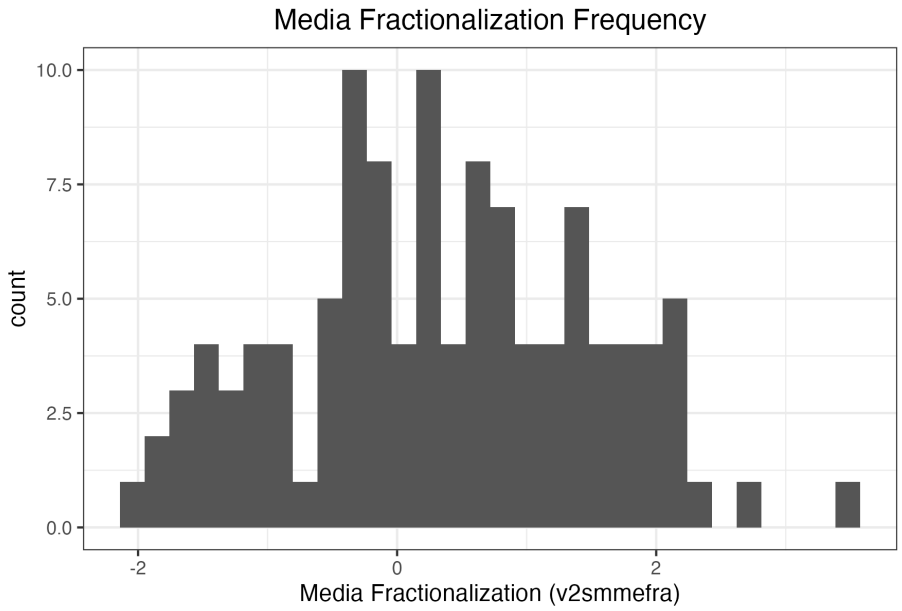
Results I: Descriptive Statistics

Variable	Description	Variable Type	Mean	Median	Mode	Standard Deviation	Interquartile Range
v2smmefra	Online media fractionalization	Interval	0.3402			1.114007	
v2cacamps	Political Polarization	Interval	-0.0688			1.411923	
v2x_regime	Regimes of the world Measure	Ordinal		2			2
e_gdppc	GDP Per Capita	Interval	20.486			19.88839	
v2mecenefi	Internet censorship effort	Interval	0.5805			1.389228	
v2expathhg	Head of Government Appointment	Nominal			6		0

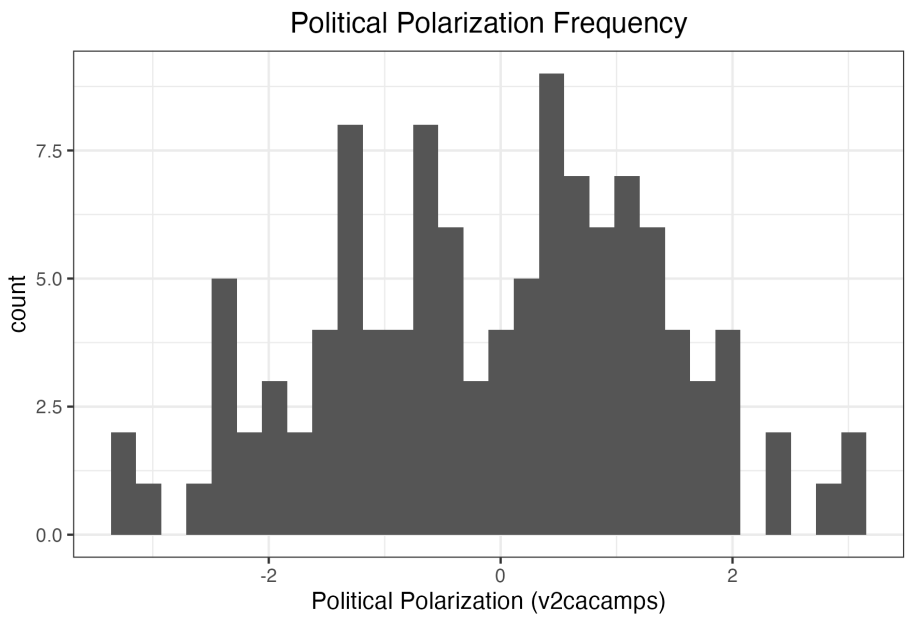
As the results of this table show, the independent variable (media fractionalization) has a mean of 0.34. This variable was initially ordinal from 0-4, but was converted to interval-level in the V-Dem dataset. Hence, the interpretation of the mean is somewhat altered, and the range is -2.04 to 3.48. A mean of 0.34 means that the arithmetic average level of fractionalization is

somewhat high and closer towards highly opposing news than highly similar news. Its standard deviation is 1.11, meaning the data is moderately dispersed. The dependent variable, political polarization, has a mean of -0.0688. Considering it ranges from -3.25 to 3.05, this mean is roughly halfway through the data, indicating the arithmetic average polarization across nations tends toward lower polarization. The regime types variable, used later in an interaction, is ordinal and has a median of 2. This means the half the data is in a ranked category above 2, and half the data is below 2. Its interquartile range is 2, meaning that the middle 50% of the data (between the 25th and 75th quartiles) spans 2 regime types.

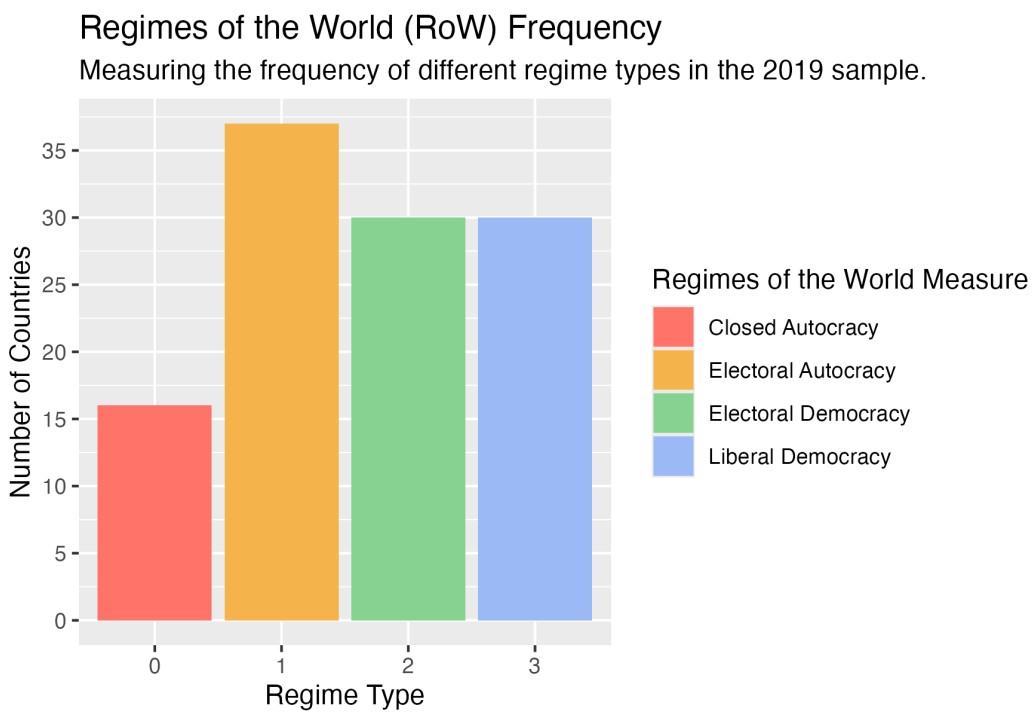
Results II: Frequencies



Next, the results of this histogram show that media fractionalization is somewhat widely dispersed, with many countries having a wide variety of media fragmentation levels. There is no singular peak the data clusters around, with much of the data being spread out widely.



As this histogram of political polarization shows, the data is widely dispersed, with two peaks around which countries are concentrated. Polarization levels are more concentrated than media fragmentation, indicating that a higher number of countries share closer polarization but vary widely in media fragmentation levels.



This bar graph essentially functions as a histogram, showing the frequency of countries falling into each regime type. Electoral autocracy is the largest category, containing 37 countries, while electoral and liberal democracies both consist of 30 countries. Closed autocracy is the smallest category, with just 12 countries. One possible drawback of the smaller number of closed autocracies is that there may be insufficient data to determine how regime type impacts the relationship between media fragmentation and political polarization.

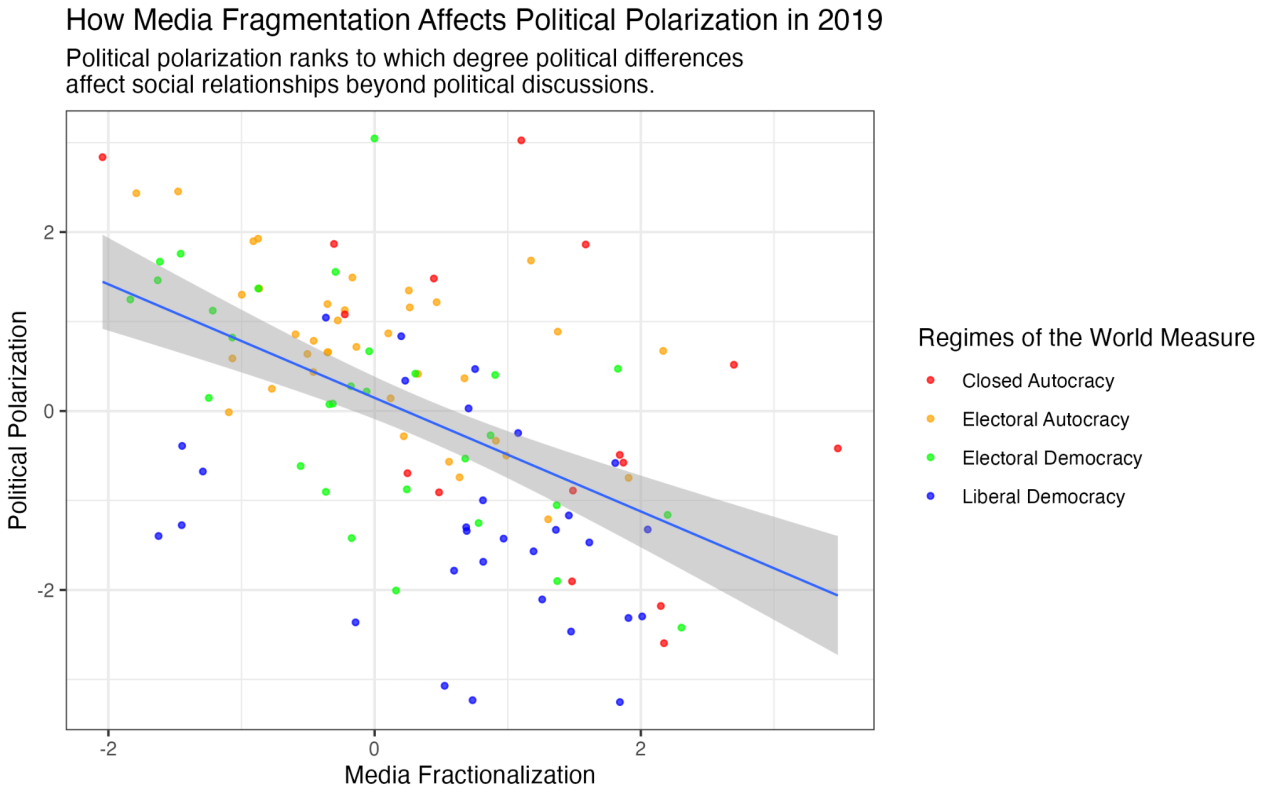
v2smmefra	*v2cacamps*				***Total***
	(-3.26,-1.68]	(-1.68,-0.104]	(-0.104,1.47]	(1.47,3.05]	
(-2.05,-0.663]	0 0 %	4 10.8 %	11 24.4 %	7 46.7 %	22 19.5 %
(-0.663,0.718]	4 25 %	12 32.4 %	28 62.2 %	5 33.3 %	49 43.4 %
(0.718,2.1]	9 56.2 %	19 51.4 %	4 8.9 %	3 20 %	35 31 %
(2.1,3.49]	3 18.8 %	2 5.4 %	2 4.4 %	0 0 %	7 6.2 %
Total	16 100 %	37 100 %	45 100 %	15 100 %	113 100 %

χ^2=38.304 · df=9 · Cramer's V=0.336 · Fisher's p=0.000

To test the hypothesis, a crosstab was created for the independent and dependent variables. Based on the results, the hypothesis appears not to be confirmed. The hypothesis is that higher media fractionalization (lower values of v2smmefra) is associated with higher political polarization (v2cacamps). The crosstab indicates that polarization is highest with mixed levels of media fractionalization, which is not aligned with the hypothesis, but could possibly indicate a non-linear relationship.

Next, a scatterplot was created to visualize how media fractionalization and political polarization are related, with a regression line to show the predicted values of the regression model.

Results IV: Data Visualization



This scatterplot plots the relationship between the independent and dependent variable, as well as the regression line with the confidence intervals. The coloured dots represent different regime types. While there is a sizeable cluster of liberal democracies low on polarization and high in similar news coverage (a high value of the media fractionalization variable), the rest of the data is quite scattered and does not visually appear to closely follow the regression line's predicted values. To further test this result, three regression models will be tested, and the coefficients, statistical significance, and R values of each are discussed.

Results V: Regression Models

Regression Model 1: No Controls

(Intercept)	0.222*
	(0.095)
Media Fractionalization	-0.659***
	(0.085)
Num.Obs.	167
R2	0.269
R2 Adj.	0.265

· p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

This regression model indicates a statistically significant relationship between media fractionalization and political polarization across the 113-country sample in 2019. For every ordinal level increase in media fractionalization (meaning higher similarity among online news coverage), there is a 0.659 unit decrease in political polarization. V-Dem converted both media fractionalization and polarization (ordinal variables) into interval using the Bayesian item response theory measurement model. This negative relationship appears to confirm my hypothesis that higher media fractionalization is correlated with higher polarization. Less similar news coverage among online news media is associated with less polarization, meaning the inverse relationship is also true. The R^2 value is 0.26, meaning the model predicts 26% of the variance in political polarization. The remaining 74% cannot be explained by the model. The second model will include all the control variables to attempt to capture more variation.

Multiple Regression Model 2: All Controls Added

(Intercept)	1.856+
	(1.060)
Media Fractionalization	-0.540***
	(0.097)
GDP per capita	-0.009
	(0.007)
Regime Type: Electoral autocracy	-0.086
	(0.370)
Regime Type: Electoral democracy	-0.650
	(0.472)
Regime Type: Liberal democracy	-1.396**
	(0.513)
Internet Censorship Effort	-0.000
	(0.124)
Head of State Appointment: By Ruling Party	-1.284
	(1.512)
Head of State Appointment: By Head of State	-0.940
	(1.101)
Head of State Appointment: By Legislature	-1.172
	(1.126)
Head of State Appointment: By Popular Election	-0.781
	(1.549)
Head of State Appointment: Other	-1.477
	(1.533)
Num.Obs.	113
R2	0.504
R2 Adj.	0.450

· $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

As this multiple regression model shows, even after including the control variables, the relationship between media fractionalization and political polarization remains statistically significant at the 95% confidence level. While the coefficient has changed from -0.659 to -0.54, this may mean the controls have helped the coefficient to become more precise as more of the variation is explained. Additionally, the liberal democracy regime type is associated with a -1.39 unit lower level of polarization, and is statistically significant at the 99% confidence level, which is more than sufficient. This is unsurprising as liberal democracies can generally promote the peaceful exchange of ideas without intense polarization. The other coefficients are not statistically significant and thus we cannot conclude anything about their impact on polarization. The R^2 value is 50%, meaning half the variation in polarization is explained by the model, which has gotten larger than in the first simple regression model. The model has become more predictive by adding control variables.

Multiple Regression Model 3: Interaction with Regime Type

(Intercept)	1.829+
	(1.067)
Media Fractionalization	-0.609**
	(0.216)
Regime Type: Electoral autocracy	-0.158
	(0.434)
Regime Type: Electoral democracy	-0.705
	(0.520)
Regime Type: Liberal democracy	-1.577**
	(0.576)
GDP per capita	-0.009
	(0.007)
Internet Censorship Effort	-0.013
	(0.126)

Head of State Appointment: By Ruling Party	-1.155
	(1.572)
Head of State Appointment: By Head of State	-0.847
	(1.141)
Head of State Appointment: By Legislature	-1.070
	(1.170)
Head of State Appointment: By Popular Election	-0.665
	(1.596)
Head of State Appointment: Other	-1.362
	(1.566)
Electoral autocracy interaction with media fractionalization	0.114
	(0.289)
Electoral democracy interaction with media fractionalization	-0.093
	(0.277)
Liberal democracy interaction with media fractionalization	0.264
	(0.287)
Num.Obs.	113
R2	0.514
R2 Adj.	0.445

· $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

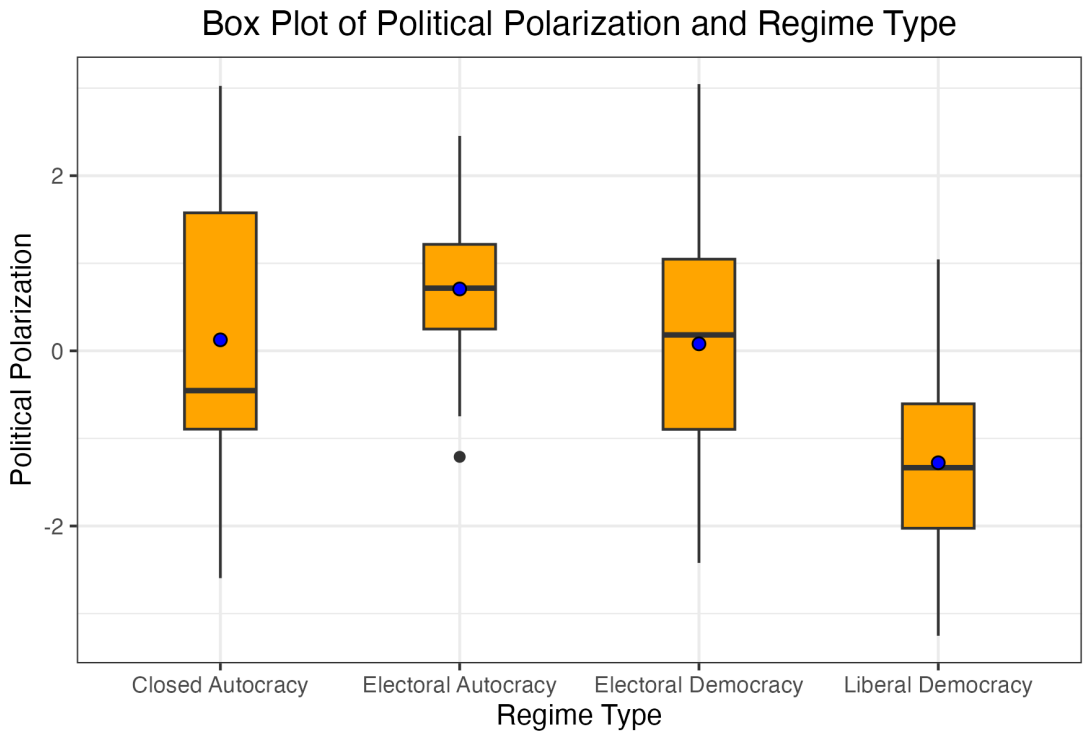
Finally, this multiple regression shows how regime type affects the impact media fragmentation has on polarization, with all the same control variables as before. Interestingly, none of the coefficients for the interaction are statistically significant, meaning there is very little that can be taken away from these results. As mentioned before, the only two statistically significant coefficients are for media fractionalization and liberal democracies. The R^2 value has increased to 51%, meaning the interaction may have helped explain 1% more of the variation in polarization. However, there is still a significant amount of variation in polarization results that cannot be explained by the model.

Discussion

In summary, this paper has explored the relationship between online media fractionalization (the proliferation of different domestic interpretations of political news), and political polarization (the level of hostility between supporters of different political camps). The main findings has been that yes, higher media fragmentation is indeed associated with higher political polarization. However, it is not possible to establish causality due to a) the use of cross-sectional data, which restricts the observation of the relationship to one particular year, 2019, and b) regression models establish correlation, not causation. It is not possible to tell with certainty whether media fragmentation directly results in higher polarization. In fact, it is entirely possible that the relationship is spurious. Reverse causality could mean that the level of polarization is driving different opinions in online news, and thus leading to higher fragmentation. Or the relationship could be so complex that both variables feed into each other in a cyclical manner. It is not possible to determine with certainty through this analysis.

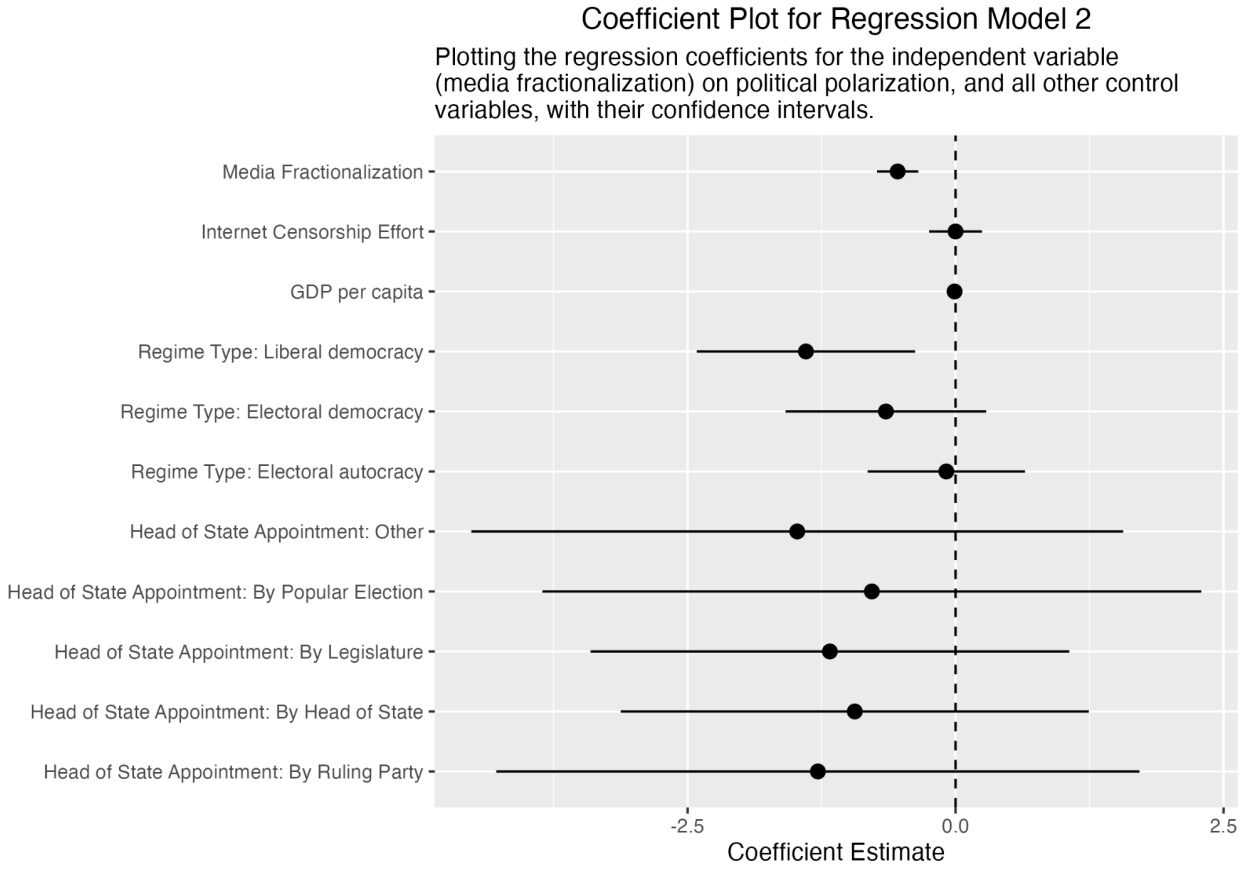
Future research should continue to study the relationship between media fragmentation and polarization, as it has important implications on economic health, state capacity, and overall government stability (Kang, Kim and Lee 2025, 1). It is recommended to pursue a research design that can determine causality, such as using panel data, which could track the same countries over time. A final important acknowledgement is that the V-Dem dataset is based on expert opinions, not actual observed data. There is a possibility for bias and human error, and as such it is possible the numbers do not reflect the real levels of polarization in these nations. However, these measures are difficult to obtain and will always be based on somewhat subjective attempts to operationalize the concepts underpinning the variables (Besco 2026, Lecture).

Appendix



In order to visualize the spread of polarization across regime types, a boxplot was created. Political polarization has the lowest mean (blue dot) in liberal and electoral democracies, which is unsurprising. This may be due to democracies emphasizing a plurality of ideas and allowing free and fair elections. It is unsurprising that liberal democracies, who take it further in allowing multiple political parties to exist, potentially leading to less social tensions and polarized feelings, have even lower mean polarization than electoral democracies, who do not fully satisfy respect for personal liberties. The median however (thick line in the boxes) shows that liberal democracies and closed autocracies actually have the lowest halfway point for polarization. This is interesting for closed autocracies, who may suppress political freedoms of association, thus crushing polarization in the 50% of data below the median (Coppedge et al.

2024, 292). As the maximums and minimums show, democracies are capable of reaching levels of polarization as high as the middle 50% of data within autocracies.



Additionally, to visualize the regression coefficients (of the second regression with all the controls and no interaction), a coefficient plot was created. This plots the coefficients as dots, and shows their respective confidence intervals. Nearly all the confidence intervals cross the 0 line, meaning within 95% confidence, there is a possibility the coefficients could be zero, indicating that no relationship between and polarization exists. This plot is a useful way to visualize the data other than the regression table. As was noted in the regression section, media fractionalization has a statistically significant relationship with polarization, and thus its confidence intervals do not cross zero (we are 95% confident the relationship between them in the population, not just the sample, is not nonexistent). The same applies to the liberal

democracy regime type: it has a statistically significant relationship with polarization that we can be 95% confident exists outside just the sample.

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